

Where Do Returns Actually Come From — Overnight or Intraday?

Martingale · Research Note 005 Author: Neil Gilani · Reproducible code: [experiment.py](#)

Abstract

Every trading day splits exactly into two sessions: the **overnight** gap from yesterday's close to today's open, and the **intraday** move from open to close. On 5,030 days of the NASDAQ Composite (1999–2018) the split is stark — overnight compounds **+917%** while the intraday session loses **−70%**, so the index's entire gain, and then some, accrued while the market was shut. The gap is **+6.42 bp/day** ($t = 2.90$, bootstrap 95% CI [+2.12, +10.78]), and a single stock with real auction prints (GOOG) replicates it. But the effect is not harvestable and not stable: break-even transaction cost is **2.46 bp per side** against the two most expensive moments of the trading day, and in the second half of the sample overnight (+7.77%) and intraday (+7.15%) are effectively **tied**. The median day is a dead heat (+6.19 bp vs +6.16 bp); what separates the sessions is that **18 of the 20 worst sessions in 20 years were intraday**. Finally, a widely distributed S&P 500 feed answers this question **backwards** because its opening prices are stale — a defect we reproduce on purpose, and give a one-line test to catch.

1. Introduction

A popular claim holds that the stock market's entire long-run return is earned overnight, and that the regular session — the part everyone watches — contributes nothing or less. Unusually for the claims this lab tests, this one is not a pattern someone spotted on a chart. It is an **accounting identity**:

$$(1 + \text{overnight}[t]) \times (1 + \text{intraday}[t]) = (1 + \text{close-to-close}[t])$$

where overnight is $\text{close}[t-1] \rightarrow \text{open}[t]$ and intraday is $\text{open}[t] \rightarrow \text{close}[t]$. Every day's return belongs to exactly one session or the other; nothing is double-counted and nothing is lost. There is no model to misspecify and no parameter to tune, which makes this one of the few questions here that can be answered exactly rather than estimated.

That leaves three questions worth asking, and this note asks all three:

1. **Is the split real** in a clean sample, and how big is it?
2. **Does it survive contact with reality** — costs, sub-periods, the removal of any single era?
3. **Is it robust to the data itself?** An identity computed on wrong inputs is still wrong.

Hypothesis (stated before running it). The overnight/intraday gap will be real and measurable — it is documented in the literature and it is an identity, not a fitted signal — but it will **not** be exploitable after realistic costs, because it requires a round trip every single day at the open and the close. A gap that survived costs, held across sub-periods, and was not concentrated in one era would refute this.

2. Data & Method

Data. Unlike Notes 002–004, which fetch live from `yfinance` and therefore produce slightly different numbers on every re-run, this note is pinned to **frozen, redistributable datasets** shipped inside published Python packages: the NASDAQ Composite and S&P 500 daily OHLC series in `arch` 8.0.0 (1999-01-04 → 2018-12-31, 5,031 bars each) and the GOOG series in `backtesting` 0.6.6 (2004-08-19 → 2013-03-01, 2,148 bars). Every number below reproduces exactly, permanently, on any machine — which for a study whose entire subject is data quality seemed like the only defensible choice.

Method. Decompose each day per the identity above, drop day 0 (it has no prior close), and assert the identity numerically rather than trusting it — the run reports a maximum residual of 2.2×10^{-16} . A "session strategy" holds the index across one session and cash across the other, so it crosses the spread **twice per day**; costs are charged accordingly. Because the two sessions are measured on the *same* days, the test of which is larger is **paired**, with a 5,000-draw bootstrap CI rather than a normal approximation on a fat-tailed series.

Validation. A self-test runs before every experiment: it checks the identity on random prices, recovers a planted overnight drift (and finds no phantom intraday drift), confirms **no** significant gap on data with no planted edge, and verifies the audit and cost accounting. All five checks pass on each run.

2.1 The data audit, run before the analysis

Because the analysis divides by the opening price, an opening price that is not real invalidates everything. Two tests run first, and their verdict decides which series may be used:

Series	Bars	Opens exactly = prior close	Overnight ÷ intraday vol	Verdict
NASDAQ Composite	5,031	0.2%	0.580	usable
GOOG	2,148	0.5%	0.787	usable
S&P 500 (same package, same window)	5,031	39.8%	0.139	rejected

The first test is blunt: some feeds carry the prior close forward as the open, and 39.8% of this S&P 500 series does exactly that. The second is the sharper one, and the reason the rejection is total rather than partial. Real equities gap on overnight news, so overnight volatility is a substantial fraction of intraday volatility — 0.58 and 0.79 in the clean series. This S&P 500 feed reports **0.139**: an annualised overnight volatility of 2.6%, and a move larger than 0.5% on just 2.2% of nights. That describes a market that does not exist. Note that the exact-match test flags only the 1999–2005 stretch, while the volatility ratio shows the series is compromised throughout — **the visible defect understates the real one**. The series is therefore excluded from the findings and reported only as the case study in §4.

3. Results

3.1 The decomposition

NASDAQ Composite, 1999–2018 (5,030 days):

Session	Mean/day	Median/day	Ann. return	Ann. vol	Sharpe	Cumulative
Overnight (close → open)	+4.93 bp	+6.19 bp	+12.32%	12.6%	+0.99	+917.4%
Intraday (open → close)	−1.49 bp	+6.16 bp	−5.93%	21.7%	−0.17	−70.5%
Buy & hold (close → close)	+3.46 bp	+8.82 bp	+5.67%	25.3%	+0.34	+200.5%

Paired difference: **+6.42 bp/day**, $t = +2.90$, bootstrap 95% CI [+2.12, +10.78] bp. Overnight was positive in 15 of 20 calendar years; intraday in 9.



GOOG, 2004–2013 (2,147 days) — an independent check on a single stock with genuine opening-auction prints, which rules out index-construction artifacts: overnight **+40.35%** annualised (Sharpe +1.69) versus intraday **−9.01%** (Sharpe −0.21); paired difference +16.62 bp/day, $t = +3.50$.

3.2 What it costs to collect

An overnight-only strategy buys the close and sells the open, every day.

Cost per side	Overnight ann. return	Sharpe
0.0 bp (gross)	+12.32%	+0.99
0.5 bp	+9.53%	+0.79
1.0 bp	+6.81%	+0.59
2.0 bp	+1.56%	+0.19
5.0 bp	-12.70%	-1.01

Break-even cost: 2.46 bp per side.

3.3 Stability

Sub-period	Overnight	Intraday
1999–2008 (first half)	+17.07%	-17.42%
2009–2018 (second half)	+7.77%	+7.15%
2001–2018 (excluding the dot-com years)	+5.72%	-0.06%

3.4 Where the gap lives

Symmetric trimmed means, in bp/day:

Trim per side	Overnight	Intraday	Gap
0% (mean)	+4.93	-1.49	+6.42
1%	+5.33	-1.70	+7.03
5%	+5.70	-0.76	+6.46
10%	+5.90	+0.94	+4.96
25%	+6.40	+4.63	+1.78

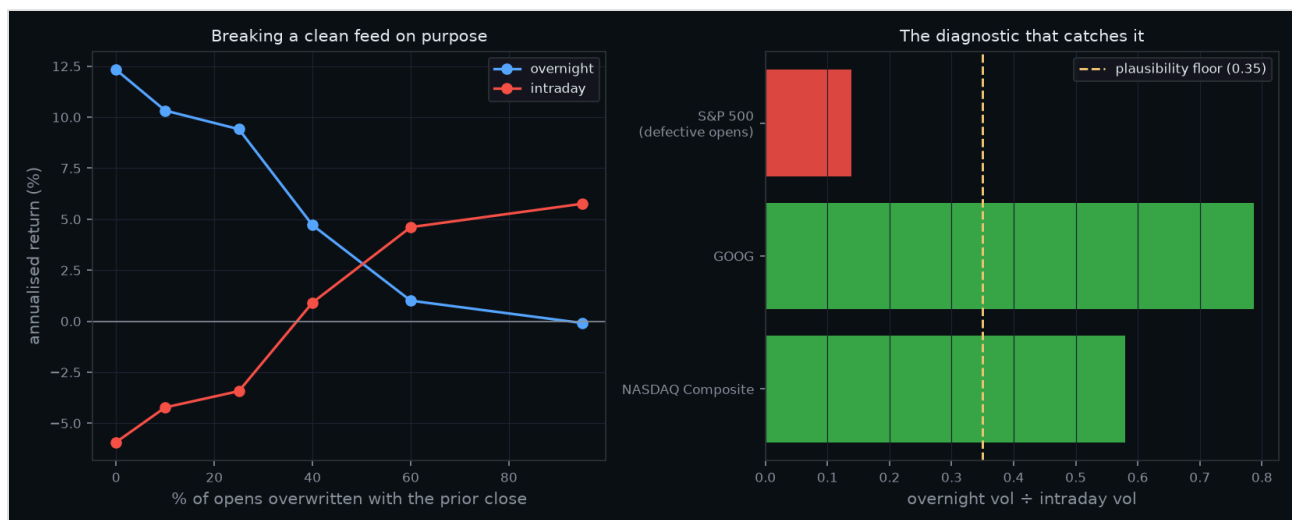
Of the **20 worst single sessions** in the twenty-year sample, **18 were intraday**.

4. The data-quality case study

Run the identical decomposition on the rejected S&P 500 feed and it returns the **opposite** answer: overnight +0.77% annualised versus intraday +2.84%, with a paired difference of -1.46 bp/day. A researcher who skipped the audit would have published "returns are earned during the session, not overnight" — with a large sample, a familiar index, a reputable package, and an exact identity underneath.

To show the defect *causes* the reversal rather than merely accompanying it, take the clean NASDAQ feed and break it on purpose: overwrite the open with the prior close on a random fraction p of days.

p (opens overwritten)	Overnight ann.	Overnight vol	Intraday ann.	Vol ratio
0% (clean)	+12.32%	12.6%	−5.93%	0.580
10%	+10.33%	12.0%	−4.22%	0.544
25%	+9.41%	10.7%	−3.42%	0.473
40%	+4.71%	9.6%	+0.91%	0.412
60%	+1.01%	7.5%	+4.61%	0.307
95%	−0.09%	2.4%	+5.76%	0.096



The mechanism is mechanical: a stale open forces that day's overnight return to exactly zero and hands the entire day to the intraday bucket. The conclusion crosses over between 25% and 40% contamination, and the answer degrades smoothly all the way — meaning **a feed can be substantially corrupted while still producing a plausible-looking, entirely wrong result**. The volatility ratio falls monotonically alongside it, which is what makes it a usable alarm.

5. Discussion

The headline is true, and it is not an edge. Over 1999–2018 the NASDAQ Composite really did earn its entire return, and more, outside regular trading hours. That is an identity, not an estimate. But collecting it means a round trip every day, at the open and the close — the two moments of the day when spreads are **widest**, precisely because that is when the auction absorbs accumulated overnight order flow. Break-even is 2.46 bp per side. At the 5 bp charged in [Note 003](#), the strategy loses 12.7% a year. The pattern is real; the trade is not. This is the same lesson as Note 003's mean reversion, arrived at from the opposite direction: there, high turnover destroyed a real regime effect; here, high turnover destroys a real accounting effect.

The dramatic version is a dot-com artifact. Of the +917% cumulative overnight return, +274.5% came from 1999–2000 alone. Drop those two years and overnight falls from +12.32% to +5.72% annualised while intraday rises from −5.93% to −0.06% — the gap shrinks by roughly two-thirds. In the second half of the sample the two sessions are effectively **tied** (+7.77% vs +7.15%). Anyone quoting the twenty-year figure as a live description of today's market is quoting the dot-com bubble.

"Returns happen overnight" is better stated as "crashes happen intraday." The median overnight day is +6.19 bp and the median intraday day is +6.16 bp — a dead heat. The typical session contributes just as much as the typical night. What differs is the left tail: intraday volatility is $1.7\times$ overnight, and 18 of the 20 worst sessions in twenty years were intraday. The trimmed-mean profile shows this is not *purely* a tail story — a gap of ~6.5 bp survives trimming 5% from each side — but it collapses to +1.78 bp at 25% trimming. The intraday session is where the market's disasters are priced; the overnight session accumulates a small, steady drift and sits out the worst of them. That is a description of **when risk is realised**, not a free lunch, and it is the economically sensible reading: whoever holds the position overnight is compensated for bearing gap risk they cannot trade out of.

The methodological finding may outlast the empirical one. Note 001 showed how a *method* lies; this one shows how *data* lies, and the failure is quieter. The S&P 500 series carries no missing values, no obvious outliers, correct OHLC ordering, and a perfectly satisfied identity. It is packaged in a well-regarded library. Nothing in a normal sanity check catches it, and the conclusion it produces is confidently backwards. The one-line defence generalises beyond this question: **compute the volatility your data implies for a quantity you already understand, and check it against physical reality.** An index whose overnight volatility is 2.6% annualised is telling you its opening prices are not real, whatever else the file looks like.

6. Limitations

Two of the three series are indices, which cannot be traded directly; the tradeable proxies (ETFs, futures) have their own costs and their own opening mechanics. GOOG is a genuine single stock with real auction prints, but it is also a spectacular ex-post winner over 2004–2013 — its overnight numbers inherit that selection, and it should be read as a check on the *sign and mechanism*, not as a return estimate. The sample ends in 2018, so it says nothing about the 2020–2026 regime; the 2009–2018 convergence suggests re-checking on recent data before repeating any twenty-year figure. Costs are modelled as a flat per-side charge, which understates the true difficulty: opening spreads are wider than the average, and an overnight strategy pays them every single day. The volatility-ratio threshold of 0.35 is a judgement call calibrated on three series — it separates these cleanly, but it is a smoke alarm, not a proof of cleanliness.

7. Conclusion

The overnight/intraday split is exactly what it is claimed to be, and almost nothing that is claimed *from* it. Across twenty years of the NASDAQ Composite the entire index gain accrued between the close and

the open, while the regular session lost money — an identity, verified to machine precision, and replicated on a single stock. It is also not tradeable at 2.46 bp of break-even cost incurred at the day's two widest spreads, it is two-thirds driven by the dot-com era, it has roughly vanished in the last decade of the sample, and at the median it does not exist at all: the sessions differ mainly in that one of them is where the crashes happen. And a widely distributed feed for the most-watched index in the world answers the question backwards, silently, unless you check the volatility its opening prices imply. The honest summary is not "buy the close, sell the open." It is that the overnight session is where the market pays you for holding risk you cannot hedge — and that you should audit your prices before you audit your strategy.

References

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